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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	SPNG
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 – Decent Work and Economic Growth
Project / Solution Name	e-Salary Guard
Name (Leader)	Dixson Oswaldo Tanjaya
Email & Phone (Leader)	dixsonoswaldooo@gmail.com | +60 1115442413
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

In most small and medium sized enterprises (SMEs), attendance, overtime, leave and salary are kept in separate places: a clock in system, chat messages, paper forms and a spreadsheet. At month end, one HR officer (often also the admin or the owner) reconciles them by hand. The results are predictable: wrong deductions, missed overtime, and employees who discover a pay cut only on payday, with no record explaining why.

The scale is significant. Malaysian MSMEs employed 8.09 million people in 2025, or 48.7% of national employment [1]. Between 2020 and 2024, the Labour Department of Peninsular Malaysia handled 28,061 wage-claim cases worth RM220 million, including 6,591 cases in 2024 alone [2]. Globally, an EY study found that one in five payrolls contains errors, that each error costs about USD 291 to correct, and that time-and-attendance and leave requests are the two most frequent error sources [3].

Lower wage and hourly workers are hit hardest: one unexplained deduction can break a monthly budget, and few feel confident enough to challenge HR. For employers, every dispute costs time, trust and staff retention. The core problem is not that deductions exist; it is that they are calculated manually, communicated late and impossible to trace.

2.2 The Proposed AI Solution

Provide a high level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

e-Salary Guard is a web application that turns attendance records and HR decisions into an accurate, explainable monthly pay figure for every employee, and notifies employees of each deduction when it happens, not on payday.

It has two role separated portals. Employees see a live, itemised estimate of this month's pay with a one sentence explanation, request overtime or leave, accept overtime offered by HR, and receive and answer WhatsApp style messages. HR sets salaries, bonuses, overtime rates per department and deduction rules, adds workers, approves requests, reviews the payroll and pays everyone with one confirmed click. Paid months are locked as snapshots, and every change is written to a read only activity log. The interface is available in English, Bahasa Melayu and Mandarin.

Where AI is used: judging unstructured text. When an employee submits a leave request, the SEA-LION large language model (Gemma SEA-LION-v3-9B-IT by AI Singapore), served on RunPod Serverless with SDC sponsor credits, reads the free text reason in Malay, English or a mix and returns a structured label, paid or unpaid leave, with a one sentence rationale and a confidence score. Labels below 0.6 confidence are flagged "Please check", the reason is treated as data rather than instructions, and a keyword fallback covers model failures or slow responses.

The AI never decides alone: HR approves, rejects or changes every label, and the label source (AI suggestion, keyword check or HR) is always shown. Pay itself is computed by deterministic, unit tested code, so every figure can be reproduced.

SDC Phase 1 skills applied: GenAI APIs (guarded structured classification), database architecture (PostgreSQL with optimistic locking) and cloud hosting (Replit deployment with server side secrets).

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- **Application:** Next.js 16 (App Router), React 19, TypeScript and Tailwind CSS in one codebase, with two role separated areas (/employee and /hr) and server side route handlers (/ai/state, /ai/session, /ai/space, /ai/classify-leave). Sign in uses a signed httpOnly cookie, and every change is checked on the server against the signed in role.
- **Database:** PostgreSQL managed by Replit. The demo state (employees, attendance, requests, payroll snapshots, messages and activity log) is stored as JSONB in one table, esalary_state, declared with Drizzle and accessed with node postgres, with a version number for optimistic locking. Each visitor gets an isolated demo space.
- **Payroll engine:** Pure TypeScript modules (lib/payroll.ts, lib/engine.ts) covered by 73 Vitest unit tests, including grace period lateness, absence, unpaid leave, overtime, locked months and negative pay. Lateness beyond a grace period is deducted at an employer set rate per minute; tax and statutory contributions are out of scope.
- **GenAI:** SEA-LION (aisingapore/Gemma-SEA-LION-v3-9B-IT) on RunPod Serverless, funded by SDC sponsor credits, called only from the server route /ai/classify-leave. The reply is parsed and checked by our own validator; a keyword fallback is used if the model fails or is slow.
- **Integrations:** Orbyte Attendance and e-Leave are replaced by synthetic data; WhatsApp and salary payments are simulated in app, ready for real connectors (e.g. WhatsApp Business Platform).
- **Hosting & DevOps:** GitHub repository deployed to Replit (Autoscale deployment); the RunPod key and database URL are stored in Replit Secrets and never reach the browser.

Data flow (see Appendix A): Browser (Employee / HR) → Next.js route handlers (session and role check) → payroll engine; leave reason text only → SEA-LION on RunPod → PostgreSQL → messages and activity log.

Sponsor resources: Yes. RunPod (SDC sponsor) credits host the SEA-LION model on serverless GPU. The AI layer is one server-side module, so another provider can be swapped in without changing the payroll engine.

3.2 Design Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we mapped the pay cycle from three viewpoints: the employee who learns of deductions on payday, the HR officer reconciling four data sources by hand, and the owner who signs off without a breakdown. Define: the core need is trust: every ringgit must be explainable and communicated early. Ideate: we cut the scope from a full HR suite to one loop: attendance → decision → pay → notice → audit. Prototype: a working web app on synthetic data that runs this loop end to end. Test: before Phase 2, at least five users (SME owners, HR staff and hourly workers) will complete timed tasks, and we will check whether employees can explain their own deductions.

Viability comes from deliberate constraints: a unit tested pay formula, AI limited to one human-reviewed task, and locked paid months. Deployment runs from GitHub to Replit with managed PostgreSQL; real Orbyte, e-leave and WhatsApp connectors can later replace the simulations through adapters.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary – Target 8.8:** "Protect labour rights and promote safe and secure working environments for all workers, including migrant workers, in particular women migrants, and those in precarious employment." Timely, transparent and traceable pay protects the most basic labour right: being paid correctly. **Secondary – Target 8.3:** "…encourage the formalization and growth of micro-, small- and medium-sized enterprises…" by digitising SME payroll records.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?*	• **1 in 5** payroll runs contain errors; time/attendance (1,139) and leave requests (721) are the largest error sources per 1,000 employees per year [3]. • **0 days** of advance notice: under manual payroll, employees first see deductions on their payslip.

Metric Category	Your Response
e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."	• **6,591** wage-claim cases (RM68.1 million) filed with the Labour Department of Peninsular Malaysia in 2024 [2]. • **~29 work-weeks** per 1,000 employees per year spent fixing the five most time consuming payroll error types [3].
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing latency improvement, or kg of resource waste avoided.*	1. **Pay accuracy:** 100% of payslips matching a hand verified answer key. 2. **Notice lead time:** 100% of deductions notified to the employee within 24 hours of the triggering event. 3. **HR processing time:** minutes per complete monthly payroll run (review to payment), app vs. spreadsheet. 4. **AI agreement:** ≥85% of leave reasons where the AI label matches the HR label.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	1. **100%** payslip accuracy on a synthetic test month covering every deduction type (late, absent, unpaid leave) and overtime (baseline: ~80% error-free runs [3]). 2. **100%** of deductions notified within 24 hours (baseline: 0%). 3. **≥70%** reduction in HR time per payroll run in timed tests with at least five users. 4. **≥85%** AI–HR label agreement on a 50-reason labelled test set, with 100% of low confidence (<0.6) cases flagged.

Sources

[1] Department of Statistics Malaysia, MSMEs Performance 2025, as reported in Business Today (30 July 2026), "MSME Sector Outpaces Broader Economy in 2025, Contributes RM689 Billion to GDP". https://www.businesstoday.com.my/2026/07/30/msme-sector-outpaces-broader-economy-in-2025-contributes-rm689-billion-to-gdp/

[2] New Straits Times (January 2026), "28,000 workers owed RM220 million in unpaid wages", citing the Labour Department of Peninsular Malaysia. https://www.nst.com.my/news/nation/2026/01/1363863/28000-workers-owed-rm220-million-unpaid-wages

[3] EY survey commissioned by Paycom (22 December 2022), "Payroll errors average $291 each, impacting the economy", Business Wire; survey of 508 U.S. payroll professionals, used here as an international proxy. https://www.businesswire.com/news/home/20221222005093/en/

SDG target wording: United Nations, Goal 8. https://sdgs.un.org/goals/goal8

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre Existing / Prior Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

No, all code created during the hackathon.

If yes, briefly describe what was pre-existing and which team member(s) owned it:

No, team member owned prior work used here.

Note: the landing page at "/" reuses the public orbyte.my page (HTML, CSS and images, owned by Orbyte / Alpha Red) to show where e-Salary Guard sits inside Orbyte; it is declared in 6.2 and is not covered by our licence.

6.2 Third Party Code, Datasets, or Models

Does your project use any third party code libraries, open source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js 16.4 (with React 19.3)	Code – framework	MIT
TypeScript 5.9	Code – language/compiler	Apache 2.0
Tailwind CSS 4.3	Code – styling library	MIT
Drizzle ORM / Drizzle Kit, node-postgres (pg) 8.23	Code – database libraries	Apache 2.0 / MIT
Vitest 5.0	Code – testing	MIT
Inter typeface (Google Fonts)	Font	SIL Open Font License 1.1
Gemma-SEA-LION-v3-9B-IT (AI Singapore)	Model – pre-trained, instruction-tuned; run on RunPod	Gemma Terms of Use (built on google/gemma-2-9b)

orbyte.my landing page (HTML/CSS/images)	Design/content	© Orbyte (Alpha Red); hackathon demo only
Synthetic employee & attendance data	Dataset – team generated	Team owned

6.3 Employment Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

Not applicable. Each member to confirm they have no such agreement.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Dixson Oswaldo Tanjaya, Team Leader, on behalf of SPNG
Date: 23 October 2026

Appendix A – System Block Diagram

Figure A1. End to end architecture of the e Salary Guard prototype: role separated portals, server side payroll engine and AI classifier, managed PostgreSQL database, and simulated external integrations.

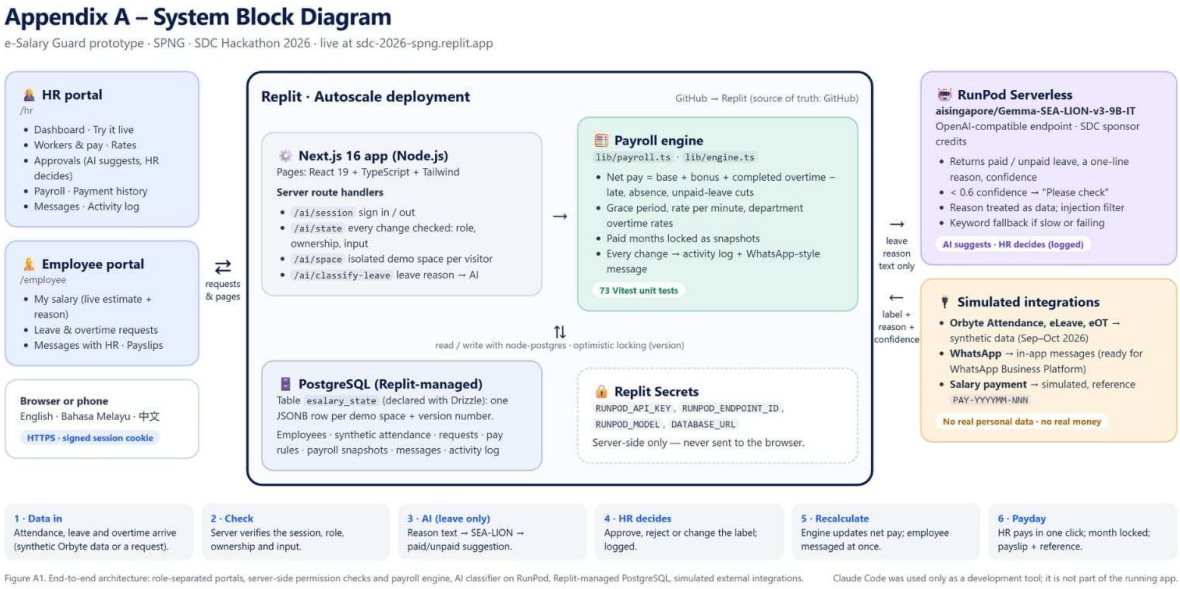


Figure A1. End-to-end architecture: role-separated portals, server-side permission checks and payroll engine, AI classifier on RunPod, Replit-managed PostgreSQL, simulated external integrations. Claude Code was used only as a development tool; it is not part of the running app.